MAGTALAS Bernie Calderon
Economic Development Specialist | Project Economics | English (Fluent) | Filipino (Native)
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Economic Development Specialist with over 8 years of experience in project economic analysis, cost-benefit analysis, infrastructure appraisal, and development project evaluation across the energy, transport, and social infrastructure sectors, assessing project alternatives, risks, development impacts, and climate resilience, in close coordination with multilateral development banks, development finance institutions, and governments

PORTFOLIO AND DATA ANALYTICS PROJECTS (R, Python, Excel)

- Portfolio Website: https://bcmagtalas.github.io/Career-Portfolio/
- Structural Equation Modeling of Global Energy Transition: https://bcmagtalas.github.io/energy-transition-thesis/
- Regional Infrastructure Development Index: https://ridi.bolt.host/
- Infrastructure Indicator Screening Dashboard: https://bcmagtalas.github.io/ridi-dashboard/

PROFESSIONAL EXPERIENCE

Department of Economy, Planning, and Development (DEPDev) **September 2016 to present**
Infrastructure Staff, Pasig City, Philippines

Supervising Economic Development Specialist **January 2025 to present**
Project Economic Analysis, Cost-Benefit Analysis, Infrastructure Appraisal, Development Project Evaluation

- Led the appraisal of the following infrastructure projects and translated findings into practical recommendations for project teams, securing Investment Coordination Committee (ICC) approval (covering economic analysis, including economic internal rate of return [EIRR] and net present value [NPV]; financial, social, and risk analysis; project alternatives and demand assumptions; greenhouse gas abatement costs; and nature-related impacts):
  - USD175 million Philippine Geothermal Resource Derisking Facility
  - USD512 million Agus-Pulangi Hydroelectric Power Plants Rehabilitation and Asset Management Project
  - USD260 million National Energy Efficiency and Conservation Program (appraisal completed; for ICC approval)
- Evaluated a USD4 million technical assistance (TA) proposal on power grid integration in Southeast Asia, estimating the economic benefits of cross-border power trade to strengthen the TA's design
- Assessed the development impact and sustainability of the KALAHI-CIDSS community-driven development project, contributing to the Asian Development Bank (ADB) project completion report

Climate Change, Environmental Sustainability, and Resilience

- Represented the Philippines in a 6-month regional knowledge exchange with senior policymakers across Southeast Asia, synthesizing best practices to mainstream climate resilience into project design and evaluation
- Led a 24-hour turnaround evaluation of proposed exemptions to the national coal moratorium, formulating recommendations on the exemptions' economic and environmental implications to inform presidential deliberation
- Analyzed the effects of the Middle East conflict on fuel and electricity supply and prices, with findings adopted in inter-agency deliberations on strengthening energy resilience through strategic petroleum reserves

Collaboration with Multilateral Development Banks, Development Finance Institutions, and Government Agencies

- Led the joint drafting with the ADB of the Terms of Reference for an infrastructure benchmarking study, initiating the evaluation of geographic and sectoral investment disparities in the Philippines
- Directed 5 ADB consultants and 5 DEPDev specialists across the energy, water, digital, transport, and social sectors to produce the Philippines' first composite measure of infrastructure development using R
- Led the consultation workshops and write-up for the Philippine Development Plan energy chapter, synthesizing inter-agency evaluations and data on energy sector outcomes
- Assessed the USD650 million Climate Change Action Program, Subprogram 3, a policy-based loan cofinanced by the Asian Infrastructure Investment Bank (AIIB), Agence Française de Développement (AFD), and the Japan International Cooperation Agency (JICA), informing loan negotiations

Project, Sector, and Economic Analysis and Data Visualization

- Automated a multi-criteria socioeconomic evaluation model in Excel to prioritize 207 infrastructure projects, optimizing resource allocation for a USD176 billion national investment pipeline
- Built an interactive web dashboard visualizing infrastructure indices and indicators computed in R and Python, adopted as a prototype for regional and sectoral assessments of infrastructure investment gaps
- Delivered urgent economic assessments of total electrification, energy subsidies, system loss, and taxation, with recommendations adopted as institutional positions before Congress and the President

Senior Economic Development Specialist **January 2020 to December 2024**

- Evaluated a USD13 billion portfolio of 12 renewable energy and 19 transportation projects, validating financial and economic models and identifying due diligence gaps to strengthen quality at entry
- Led the risk assessment for 8 Public-Private Partnership (PPP) transportation projects, integrating risk mitigation strategies and costs into PPP contracts worth USD10 billion
- Mapped the Indo-Pacific Economic Framework (IPEF) Clean Economy Agreement against national climate targets, supporting policy alignment during inter-ministerial negotiations on regional climate cooperation
- Applied comparative policy analysis to benchmark the Natural Gas Industry Development Act against the electric power industry's market framework, formulating recommendations adopted in legislative deliberations
- Led the formulation of statutory provisions aligning the Electric Vehicle Industry Development Act with the National Transport Policy, adopted in the enacted implementing rules and regulations

Economic Development Specialist **September 2016 to December 2019**

- Estimated the economic costs of alternative energy sources using energy consumption and investment data across 17 regions, setting the benchmark for economic evaluation of energy projects
- Directed the procurement and implementation of USD2 million in consulting contracts for 14 master plans and feasibility studies on renewable energy and sustainable infrastructure, guiding regional investment programming

Department of Energy (DOE), Taguig City, Philippines **December 2015 to August 2016**
Science Research Specialist

- Collaborated with 4 World Bank consultants on field evaluations of solar projects in 7 remote areas, identifying rural electrification bottlenecks and strategies integrated into the Access to Sustainable Energy Program

EDUCATION

Waseda University, Tokyo, Japan **September 2022 to September 2024**
Master of Arts in International Relations (Specialization: Economic Development and Policy in Asia)

- **Scholarship:** Fully funded JICA Sustainable Development Goals (SDGs) Global Leader Program
- **Thesis:** Built confirmatory factor analysis and structural equation models in R to analyze 27 variables across 120 countries, measuring the effects of political, economic, and environmental factors on global energy transition
- **Relevant Coursework:** Analysis of Quantitative Data I and II, Multivariate Statistics, Sustainable Development and International Development Cooperation, Data Science for Management, and Environmental Economics and Environmental Policy

University of the Philippines Los Baños, Laguna, Philippines **May 2008 to August 2014**
Bachelor of Science in Electrical Engineering (Major: Power Engineering)

GLOBAL LEADERSHIP

World Energy Council (WEC), London, United Kingdom (Remote) **March 2023 to April 2024**
Congress Future Energy Leader

- Collaborated with 65+ Future Energy Leaders as a regional commentary contributor for Asia, informing the 2024 World Energy Trilemma Report adopted by 3,000+ member organizations of the WEC

CERTIFICATIONS

- **Energy and Flexibility Modeling (OSeMOSYS & FlexTool):** Climate Compatible Growth, June 2025
- **Modeling, Policy and Political Economy:** Climate Compatible Growth, May 2025
- **PPP Professional (CP3P), Foundation:** APMG International, January 2021
- **Green Finance Specialist:** Renewables Academy, November 2017
- **Registered Electrical Engineer:** Professional Regulation Commission, April 2015